Title:

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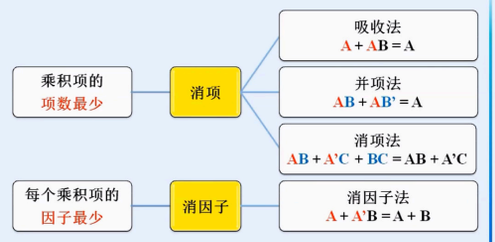
逻辑函数化简 — 公式化简 / 卡诺图

最简与或形式：乘积项项数最少；每个乘积项的因子最少

$Y = A'BC + AB'C + ABC'$

十一、公式化简法

- 1、公式化简法思路



2、吸收法 $A + AB = A$

$$Y_1 = ((A'B)'+C)ABD + AD$$
$$Y_1 = AD((A'B)'+C)B + AD$$
$$Y_1 = AD$$

$$Y_2 = AB + ABC' + ABD + AB(C'+D')$$
$$Y_2 = AB + ABC' + ABD + AB(C'+D')$$
$$Y_2 = AB$$

3、并项法 $AB + AB' = A$

$$Y_1 = A(B'CD)' + AB'CD$$
$$Y_1 = A(B'CD)' + AB'CD$$
$$Y_1 = A$$

$$Y_2 = AB' + ACD + A'B' + A'CD$$
$$Y_2 = A(B'+CD) + A'(B'+CD)$$

4、消项法 $AB + A'C + BC = AB + A'C$

$$Y_1 = AC + AB' + (B+C)'$$
$$Y_1 = AC + AB' + B'C'$$
$$Y_1 = AC + B'C' + AB'$$
$$Y_1 = AC + B'C'$$

$$Y_2 = AB'CD' + (AB')'E + A'CD$$
$$Y_2 = AB'CD' + (AB')'E + A'CD$$
$$Y_2 = AB'CD' + (AB')'E$$

5、消因子法 $A + A'B = A + B$

$$Y_1 = B' + ABC$$
$$Y_1 = B' + BAC$$
$$Y_1 = B' + AC$$

$$Y_2 = AB' + B + A'$$
$$Y_2 = AB' + B + A'$$
$$Y_2 = B + A + A'B$$
$$Y_2 = B + A + A'B$$
$$Y_2 = A + B$$

(A) + (B) + (B) = (A) + (B)

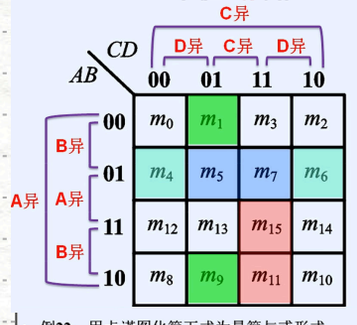
6、配项法

利用 $A + A = A$，重复写入某一项

$$Y_1 = A'BC' + A'BC + ABC$$
$$Y_1 = A'BC' + A'BC + A'BC + ABC$$
$$Y_1 = A'B + BC$$

利用 $A + A' = 1$，在某一项上乘以 $A + A'$

$$Y_2 = AB' + A'B + BC' + B'C$$
$$Y_2 = AB' + A'B(C+C') + BC' + (A+A')B'C$$
$$Y_2 = AB' + A'BC + A'BC' + BC' + AB'C + A'B'C$$
$$Y_2 = AB' + AB'C + BC' + A'BC' + A'CB + A'CB'$$
$$Y_2 = AB' + BC' + A'C$$

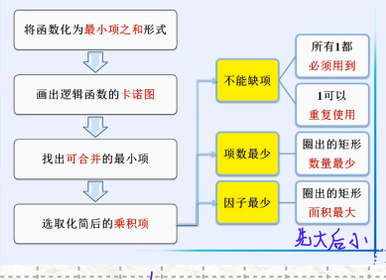


左右相邻

上下相邻

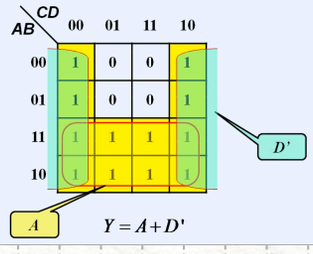
左右 + 上下 = 四角相邻

- 5、卡诺图化简逻辑函数



- 例22：用卡诺图化简下式为最简与或形式

$$Y = ABC + ABD + AC'D + C'D' + AB'C + A'CD'$$



$Y = A + D'$

8 4 2 1

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